LESSON GUIDE

Technology Plus

Unit 5: Software

3.5.2 – Generative AI

Lesson Overview:

In this lesson, students will explore how generative AI works and what it can create. Students will identify common uses of generative AI, understand the concept of prompts, and examine both the possibilities and limitations of this technology. During the lab, students will use Microsoft Copilot in the CYBER.ORG Range to troubleshoot broken HTML and personalize a web page – even if they have no prior coding experience.

Students will:

- Describe what generative AI is and give examples of what it can create.
- Use Copilot to fix or customize a basic HTML file with AI support.
- Reflect on how AI can support creativity, even when you're not an expert.

Guiding Question: How can AI help people create new things like code, images, or text – even if they don't know how to do it themselves?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.5 – Identify common uses of artificial intelligence (AI)
  - AI chatbots
  - AI assistants
  - Generative AI
    - AI-generated code
    - AI-generated content
  - AI predictions and suggestions

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).

Lesson Background

Background Information:

Generative AI tools have rapidly expanded across industries and education. These tools don't just automate tasks – they create new content based on a user's input (called a prompt). Students encounter generative AI in creative apps, productivity tools, coding assistants, and even entertainment. However, these tools have limits. They can make mistakes, repeat bias, or produce convincing but incorrect information. This lesson encourages students to understand generative AI as a powerful tool that should be used thoughtfully.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.5.2 – Generative AI • Lab Slides 3.5.2 – Prompt to Page: What Can AI Do for You?
Materials per Student	• Student Note 3.5.2 – Generative AI • Student Handout 3.5.2 – Generative AI • Printer/Copier paper • Scissors • Tape • Ruler (or other straight edge) • Assessment 3.5.2 – Generative AI (optional)

Lesson Background

Cyber Terms:

Term	Definition
generative AI	artificial intelligence that makes new content like text, images, or code
prompt	the instructions or question you give the AI
AI-generated content	text, art, or other media made by artificial intelligence
AI-generated code	code written by artificial intelligence based on your prompt
AI drift	when artificial intelligence starts performing worse over time because of updates or learning from bad input

3.5.2 – Generative AI

Guiding Question: How can AI help people create new things like code, images, or text – even if they don't know how to do it themselves?

Lesson Launch (15-20 minutes)

- Ask students how they learned to do things before AI. Answers will probably include learning things from parents or friends, reading about things, and watching videos.
- Survey the room on how many students feel like they are good at origami as you give students one sheet of printer paper (colored is more fun), a pair of scissors, and a ruler or some other straight edge. They will also need a pencil.
- Tell students that you are going to show them a video on how to make a paper fidget toy. Show the video, pausing and backing up as necessary for students to successfully create their fidget toy. Offer small pieces of tape to students to keep their seams together once the toy is complete.
- Questions for students:
    - Did you consider this activity difficult?
    - Did the video help make it easier?
    - Do you think anyone could make a fidget toy like this with enough time or the right instructions?
    - What if it was something harder – like writing a song, building a website, or coding an app? Would a video still work, or would you need something more interactive… like an AI you can talk to?
- Tell students today's big idea is this: You were one video away from building a fidget toy. Now, with generative AI, you're often just one well-written prompt away from making something much more powerful – even if you don't know how to do it yet!

Generating Buzz for Generative AI (15-20 minutes)

- Distribute Student Handout 3.5.2 – Generative AI and encourage students to complete the handout as you go through today's lesson.
- Define ***generative AI*** and ***prompts***.
- Highlight ***AI-generated content*** types and the tools AI uses to create them (text, images, code, etc.).
- Optional prompt: "What kind of AI do you use in your day-to-day life and what do you use it for?"
- Show the haiku prompt example and discuss creativity.
- Optional prompts:
    - "If you saw a poem – or image based on the poem – like this online would you think a human made it or AI?"
    - What kinds of things have you seen online that were made by AI? What do you like about them? Is there anything that gives you pause about what people are able to create?"

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- Introduce ***AI-generated code*** and explain that it doesn't understand, it predicts.
- Optional prompt: "Why might AI be helpful for someone learning to code – or someone who doesn't know how to code at all?"
- Cover benefits (speed, creativity, support) and limitations (hallucinations, bad data, ***AI drift***) of generative AI.
- Optional prompt: "Which of these is/would be the reason you turn to generative AI for a project?"
- Read each AI limitation aloud and connect it to a real-world risk:
  - Lack of understanding: It might give wrong answers that sound confident, which can be confusing or misleading in schoolwork, healthcare, or legal advice.
  - Making things up: It could invent facts, fake quotes, or make up links – imagine using one in a report and getting called out for spreading false info.
  - Learning from bad information: If the data is biased or false, AI might give unfair, offensive, or dangerous recommendations – like rejecting a job applicant based on their name or giving bad medical info.
- Optional prompt: "AI can't verify facts. What happens if it gives you an answer that sounds right – but isn't?"
- Share usage tips: Be specific, fact-check, don't overshare.
- Optional prompt: "If AI tools are already a part of your life, what's something you want to use more carefully or creatively?"
- End with the importance of using AI as a tool – not a replacement for human thinking.

Lab: Prompt to Page: What Can AI Do for You? (30-45 minutes)

- Display Lab Slides 3.5.2 – Prompt to Page: What Can AI Do for You and have students record their answers in the handout.
- Emphasize that students do not need to know HTML to complete the lab successfully – AI is their support tool.
- Walk through the lab context slide to frame the activity as an exploration, not a test.
- The lab is split into two parts:
  - Part 1: What's Wrong Here?

    Students paste/type broken code snippets into Copilot and summarize what it tells them.

    Encourage students to ask clarifying follow-up questions to Copilot if its answers are unclear or vague.
  - Part 2: Markup Makeover

    Students open the provided HTML file (myWebsite) and use Copilot to:

    - Fix errors.

- Make personal changes (text, images, colors).
- Add new HTML elements based on green comments.

- If students don't understand what the code does, prompt them to ask Copilot: "What does this HTML code do?"
- Circulate to help troubleshoot issues like:
  - File saving.
  - Image preview problems.
- Remind students that images may not display, but alt text will still show up – this is expected behavior.
- Decide how you want to receive a copy of their website… or if you do. The range is not persistent, so once the machines are terminated, student work is lost forever. You could explore:
  - Having the students log into your LMS while in the range to submit their HTML code as an assignment.
  - Logging into any school-assigned cloud storage to save a copy of their file there as well.
- Students can also copy and paste code out of the range by using the clipboard by pressing CTRL + ALT + SHIFT and then pasting the code into an app on their actual machine like Notepad or a Google Doc.
- Wrap up the lab with the Putting It All Together slide – students should reflect on:
  - How did Copilot help?
  - What surprised them?
  - How might they use generative AI again?

Putting It All Together (5 minutes)

- What makes generative AI different from other types of AI?

  Other types of AI make decisions, predictions, or suggestions based on patterns or scripts. Generative AI can create based on the prompt, or question, you give it.
- What are some real examples of things generative AI can create?

  Generative AI can create essays, poems, resumes, images, videos, music, or code.
- What's one risk or limitation of using generative AI?

  Student answers may vary but could include that generative AI might sound confident in its answers but still be wrong.

- Why is it important to know how generative AI works – even if you're not a programmer or artist?

    Student answers may vary but include that you don't have to be a programmer or an artist to use generative AI. If you have access and a question, generative AI can help you do things or learn how to do things you didn't know how to do before.

- Generative AI isn't just a tech trend – it's a new way people are creating, coding, and solving problems. From writing essays to making music, it opens up exciting possibilities. But it also comes with risks, like spreading bad info or copying others' work without credit. The more you understand how it works (and where it can go wrong), the better choices you can make when using generative AI. Think of it like a superpowered tool – it's impressive, but it still needs a smart human in charge